Dr. Pavan Kumar Meena

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Research Profile

Geotechnical Earthquake Engineer and Postdoctoral Senior Project Scientist at IIT Mandi, driving international advancements in multi-hazard resilient infrastructure. My research bridges advanced theoretical mechanics and global policy frameworks, supported by a Ph.D. from IIT Roorkee that pioneered seismic stability analyses for geosynthetic-reinforced soil slopes under complex tension crack and pore water pressure dynamics. Currently, I leverage this expertise within a high-profile global academic consortium including the University of Canterbury (NZ) and Durham University (UK) to design the Primer on Disaster Resilient Infrastructure (PoDRI), aiming to standardize global engineering practices for disaster risk reduction.

Research Interests

Slope Stability Analysis, Earthquake Geotechnics, Soil Dynamics, Ground Improvement, Ground Response Analysis.

Education

Senior Project Scientist (Postdoctoral Researcher) Sep 2026 – Present
School of Civil and Environmental Engineering
Indian Institute of Technology (IIT) Mandi

Supervisor Team: Dr. Ashutosh Kumar (IIT Mandi)
Dr. Ellen Robson (University of Canterbury, NZ)
Prof Bruce Malamud (Durham University, UK)

Doctor of Philosophy (Ph.D.) – Geotechnical Engineering Feb 2021 – Aug 2026
Indian Institute of Technology (IIT) Roorkee

Thesis Title: Dynamic Stability Analysis of Reinforced Soil Slope Considering Tension Cracks
Supervisor: Dr. Kaustav Chatterjee

Master of Technology (M.Tech.) – Geotechnical Engineering Jul 2017 – Sep 2019
National Institute of Technology (NIT) Kurukshetra

Thesis Title: A Study on Black Cotton Soil Reinforced with Nylon Fibers
Supervisor: Prof. Ashwani Jain

Bachelor of Technology (B.Tech.) – Civil Engineering Jul 2010 – Jun 2014
Rajasthan Technical University

Publications

Journal Papers

1. Meena, P. K. and Chatterjee, K. (2025). "A limit equilibrium based pseudo-static approach to investigate the role of tension cracks and water forces on internal stability of geosynthetic reinforced soil slopes." *Geotechnical and Geological Engineering (Springer)*, 43(409). DOI: 10.1007/s10706-025-03387-6

2. Meena, P. K. and Chatterjee, K. (2024). "Effect of tension crack formation on external seismic stability analysis of geosynthetic reinforced soil slopes." *International Journal of Geomechanics (ASCE)*, 24(6). DOI: 10.1061/IJGNAI.GMENG-9092

3. Meena, P. K. and Chatterjee, K. (2026). "strain dependent ground response analysis and its effect on dynamic internal stability of reinforced soil slopes with tension cracks." *Soil Dynamics and Earthquake Engineering, Elsevier. (Under Review)*

4. Meena, P. K. and Chatterjee, K. (2026). "A strain dependent modified pseudo-dynamic based approach for external stability analysis of reinforced soil slope considering tension cracks." *Natural Hazards Review, ASCE. (Under Review)*

Conference Papers

1. Meena, P. K. and Chatterjee, K. (2026). "Internal stability analysis of geosynthetic reinforced steep soil slopes considering the effect of tension cracks", *Proceedings of 17th IACMAG Conference, Hong Kong*. DOI: 10.1007/978-981-95-5969-5_26

2. Meena, P. K. and Chatterjee, K. (2023). "Influence of pore pressure parameter on seismic stability analysis of geosynthetic reinforced soil slope", *Proceedings of 17th Asian Regional Conference on Soil Mechanics (17ARC), Kazakhstan*. DOI: 10.1201/9781003299127-158

3. Meena, P. K. and Jain A. (2019). "Swell Behaviour of an Expansive Soil Reinforced with Randomly Oriented and Distributed Nylon Fiber", *National Conference on Advances in Building-Road Materials and Construction Engineering, MNIT Jaipur*.

4. Meena, P. K. and Jain A. (2018). "Improvement in Engineering Behaviour of Expansive Soils Treated with Discrete Fibers", *National Conference on Advanced Structures, Materials and Methodology in Civil Engineering, NIT Jalandhar*.

Conference Participation

- Attended and presented a paper at the 17th Asian Regional Conference on Soil Mechanics and Geotechnical Engineering (17ARC), Astana, Kazakhstan, August 14–18, 2023.
- Attended the 8th International Conference on Earthquake Geotechnical Engineering (8ICEGE 2024), Osaka, Japan, May 7–10, 2024.
- Attended and presented a paper at the 17th International Conference of the International Association for Computer Methods and Advances in Geomechanics (IACMAG 2025), Hong Kong, December 18–21, 2025.

Technical Skills

MATLAB, PLAXIS 2D, ABAQUS CAE, Geo5, AutoCAD, Civil 3D, Microsoft Office

Internship

Industrial Trainee – North Western Railway

Worked on construction of Railway Under Bridge (RUB), Jodhpur. Jun 2013 – Jul 2013

Position of Responsibilities

- Member of International Society for Soil Mechanics and Geotechnical Engineering (ISSMGE).
- Student member of American Society of Civil Engineers (ASCE).
- Lifetime member of Indian Geotechnical Society.

Award and Achievement

- Received funding from International Society for Soil Mechanics and Geotechnical Engineering (ISSMGE) Foundation to attend an international conference "17th IACMAG Conference, Hong Kong" in 2026.

- Received funding from Dean of Resources and Alumni Affairs (DORA), IIT Roorkee to attend an international conference "17th Asian Regional Conference on Soil Mechanics (17ARC), Kazakhstan" in 2023.

- Qualified GATE 2017 in Civil Engineering.

Languages Known

Hindi, English